

Rithwak Somepalli

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

B.S. in Computer Science

Expected Graduation, May 2027

Related Coursework: Data Structures, Computer Organization & Programming, Artificial Intelligence, Statistics for CS

EXPERIENCE

Twin Brain, Evu

Remote

Software Development Engineering Intern

Jun 2025 – Aug 2025

- Engineered high-throughput data pipelines using Python and Java to consume, normalize, and aggregate real-time market data from 3+ diverse brokerage APIs, processing over 5GB of daily data with 99.9% accuracy.
- Collaborated with frontend teams to architect low-latency WebSocket streams, reducing dashboard data lag by 40% (from 500ms to <300ms) and enabling instant price updates.
- Implemented robust error handling and rate-limiting strategies, ensuring 99.99% system availability during effective usage spikes of 10k+ requests/minute.

PROJECTS

Heart Anomaly Detection

Technologies: PyTorch, Kubernetes, FastAPI, WebSockets, Docker, CI/CD

Dec 2025

- Architected a resilient microservices deployment on Kubernetes, leveraging Horizontal Pod Autoscaling (HPA) to scale from 1 to 10 replicas in under 30 seconds to handle burst loads.
- Engineered an "Early Exit" Progressive Neural Network in PyTorch, utilizing entropy-based thresholds to reduce average inference latency by 35% while maintaining 95%+ accuracy.

Cloud-Native Infrastructure & DevOps Pipeline

Technologies: Terraform, OCI, Python, Bash, IAM

Nov 2025

- Architected a production-grade Infrastructure as Code (IaC) deployment on Oracle Cloud using Terraform to automate network and compute provisioning, slashing manual setup time from 1 hour to <5 minutes (90%+ reduction).
- Engineered a Zero-Trust security model utilizing IAM Instance Principals for keyless access, protected by a custom Python/Flask API with Bearer Token Authentication.
- Developed a self-healing disaster recovery pipeline using OCI Object Storage and Bash/Cron, ensuring 99.9% service availability through automated backups.

Co-Evolutionary Market Simulator (DRL vs EGT)

Technologies Python, TensorFlow, Keras, NumPy

Aug - Oct 2025

- Built a complex co-evolutionary market simulation where a DDQN Market Maker trades against a population of 50+ EGT agents over 100,000+ training episodes.
- Designed a "Shape-Shifting" hybrid reward function to train the ARL agent on 5 years of real OHLCV data, achieving a 15% higher Sharpe Ratio than the baseline strategy.

Network Intrusion Detection System

Technologies: Node.js, OCI, Linux (Iptables), REST APIs, Bash

Sept 2025

- Built a distributed, cloud-native NIDS capable of inspecting 1,000+ packets/second across a distributed network of local honeypots.
- Implemented custom TCP/UDP packet handling in Node.js to detect 5+ distinct attack types (SQLi, XSS, etc.) with a false positive rate under 2%.

SKILLS

Programming: Python, JavaScript, Java, C++, Bash/Shell, SQL, HTML/CSS, HCL (Terraform)

Frameworks & Libraries: PyTorch, TensorFlow, Keras, FastAPI, ReactJS, Node.js, Flask, Pandas, NumPy, Scikit-learn

DevOps & Cloud: Kubernetes, Docker, Terraform, Oracle Cloud (OCI), GitHub Actions, Linux (Systemd, Iptables), Git

Concepts: Microservices, REST APIs, CI/CD, Reinforcement Learning, Network Security, Zero-Trust Architecture, Game Theory

CERTIFICATIONS

Cybersecurity Certification

Google, May 2023